

# The Deficiency Indices of $D$

Diego Rincón

Independent

## Abstract

Let  $D = -i d/dt$  on  $L^2(\mathbb{R}, dt)$ . We prove that  $D$  is essentially self-adjoint with deficiency indices  $(n_+, n_-) = (0, 0)$ . This is the complete proof for the full real line. We then state precisely what must be verified to transfer this result to the adèle class space  $X = \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^*$ , where self-adjointness of  $D$  implies the Riemann Hypothesis.

## 1 Setup

**Definition 1** (The operator  $D$  on  $L^2(\mathbb{R})$ ). Let  $\mathcal{H} = L^2(\mathbb{R}, dt)$  with inner product  $\langle f, g \rangle = \int_{-\infty}^{\infty} f(t) \overline{g(t)} dt$ . Define the operator  $D$  with domain  $\mathcal{D}(D) = C_c^\infty(\mathbb{R})$  by

$$Df = -i \frac{df}{dt}.$$

**Definition 2** (Deficiency indices). For a symmetric operator  $D$  on  $\mathcal{H}$ , the *deficiency subspaces* are

$$\mathcal{K}_{\pm} = \ker(D^* \mp i) = \{f \in \mathcal{D}(D^*) : D^* f = \pm i f\},$$

and the *deficiency indices* are  $n_{\pm} = \dim \mathcal{K}_{\pm}$ . By von Neumann's theorem:

- $n_+ = n_- = 0$ :  $D$  is essentially self-adjoint (unique self-adjoint closure).
- $n_+ = n_- = k > 0$ :  $D$  has a  $k^2$ -parameter family of self-adjoint extensions.
- $n_+ \neq n_-$ :  $D$  has no self-adjoint extension.

## 2 The Full Line: Complete Proof

**Lemma 3** (Symmetry of  $D$ ).  $D$  is symmetric on  $C_c^\infty(\mathbb{R})$ : for all  $f, g \in C_c^\infty(\mathbb{R})$ ,

$$\langle Df, g \rangle = \langle f, Dg \rangle.$$

*Proof.* Integrate by parts:

$$\langle Df, g \rangle = \int_{-\infty}^{\infty} (-i f'(t)) \overline{g(t)} dt = -i \left[ f(t) \overline{g(t)} \right]_{-\infty}^{\infty} + i \int_{-\infty}^{\infty} f(t) \overline{g'(t)} dt.$$

Since  $f, g \in C_c^\infty(\mathbb{R})$ , the boundary term vanishes. Therefore

$$\langle Df, g \rangle = i \int_{-\infty}^{\infty} f(t) \overline{g'(t)} dt = \int_{-\infty}^{\infty} f(t) \overline{(-i g'(t))} dt = \langle f, Dg \rangle. \quad \square$$

**Lemma 4** (Maximal domain of  $D^*$ ). *The adjoint  $D^*$  has domain  $\mathcal{D}(D^*) = H^1(\mathbb{R}) = \{f \in L^2(\mathbb{R}) : f' \in L^2(\mathbb{R})\}$ , and  $D^*f = -if'$  on this domain.*

*Proof.* By definition,  $g \in \mathcal{D}(D^*)$  if and only if  $f \mapsto \langle Df, g \rangle$  is continuous on  $\mathcal{H}$ . For  $f \in C_c^\infty(\mathbb{R})$ :  $\langle Df, g \rangle = \int (-if')\bar{g} dt = \int f' \overline{(ig)} dt$ . The distributional derivative of  $g$  satisfies  $\int f'\bar{g} dt = -\int f\bar{g}' dt$  for  $g \in H^1$ , giving  $\langle Df, g \rangle = \langle f, -ig' \rangle = \langle f, D^*g \rangle$ . The maximal domain on which this distributional formula defines an  $L^2$  operator is precisely  $H^1(\mathbb{R})$ .  $\square$

**Theorem 5** (Essential self-adjointness on  $\mathbb{R}$ ).  *$D = -id/dt$  on  $C_c^\infty(\mathbb{R})$  is essentially self-adjoint. Its unique self-adjoint closure is  $\bar{D} = D^*|_{H^1(\mathbb{R})}$ . The deficiency indices are  $(n_+, n_-) = (0, 0)$ .*

*Proof.* We show  $\mathcal{K}_+ = \mathcal{K}_- = \{0\}$ .

**Step 1: Solve  $D^*f = if$ .** The equation  $D^*f = if$  is  $-if' = if$ , i.e.,  $f' = -f$ . The general solution is  $f(t) = Ce^{-t}$  for  $C \in \mathbb{C}$ . Check  $L^2$ :

$$\|f\|^2 = |C|^2 \int_{-\infty}^{\infty} e^{-2t} dt = \infty,$$

since  $e^{-2t} \rightarrow \infty$  as  $t \rightarrow -\infty$ . Therefore  $f \notin L^2(\mathbb{R})$ , so  $\mathcal{K}_+ = \{0\}$  and  $n_+ = 0$ .

**Step 2: Solve  $D^*f = -if$ .** The equation  $-if' = -if$  gives  $f' = f$ . The general solution is  $f(t) = Ce^t$ . Check  $L^2$ :

$$\|f\|^2 = |C|^2 \int_{-\infty}^{\infty} e^{2t} dt = \infty,$$

since  $e^{2t} \rightarrow \infty$  as  $t \rightarrow +\infty$ . Therefore  $f \notin L^2(\mathbb{R})$ , so  $\mathcal{K}_- = \{0\}$  and  $n_- = 0$ .

Since  $n_+ = n_- = 0$ ,  $D$  is essentially self-adjoint by von Neumann's theorem.  $\square$

**Corollary 6** (Real spectrum). *The self-adjoint closure  $\bar{D}$  of  $D$  on  $L^2(\mathbb{R})$  has real spectrum: all eigenvalues  $\lambda \in \mathbb{R}$ .*

*Proof.* Immediate from the spectral theorem for self-adjoint operators.  $\square$

### 3 Contrast: The Half-Line Fails

**Proposition 7** (Essential self-adjointness fails on  $(0, \infty)$ ). *Let  $D = -id/dt$  on  $C_c^\infty(0, \infty) \subset L^2(0, \infty)$ . The deficiency indices are  $(n_+, n_-) = (1, 0)$ .  $D$  has no self-adjoint extension.*

*Proof.*  $n_+ = 1$ : The equation  $D^*f = if$  gives  $f(t) = Ce^{-t}$ . On  $(0, \infty)$ :  $\int_0^\infty e^{-2t} dt = \frac{1}{2} < \infty$ . So  $e^{-t} \in L^2(0, \infty)$  and  $n_+ = 1$ .

$n_- = 0$ : The equation  $D^*f = -if$  gives  $f(t) = Ce^t$ . On  $(0, \infty)$ :  $\int_0^\infty e^{2t} dt = \infty$ . So  $e^t \notin L^2(0, \infty)$  and  $n_- = 0$ .

Since  $n_+ \neq n_-$ ,  $D$  has no self-adjoint extension.  $\square$

*Remark 8* (Why the full line matters). The contrast between Theorem 5 and Proposition 7 is the key structural fact.  $D$  on the full line is self-adjoint.  $D$  on the half-line is not and cannot be made so. The adèle class space argument for RH requires that the effective domain of  $D$  is the full line, not a half-line.

## 4 Transfer to the Adèle Class Space

**Definition 9** (The adèlic operator). Let  $X = \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^*$  be the adèle class space,  $d^*x$  the multiplicative Haar measure on  $X$ , and  $\mathcal{H} = L^2(X, d^*x)$ . The scaling action of  $\lambda \in \mathbb{R}_+^*$  on  $X$  is  $(U(\lambda)f)(x) = f(\lambda^{-1}x)$ . Under the coordinate  $t = \log |\lambda|$ , this is translation. The generator is  $D = -i d/dt$  acting via this coordinate.

**Conjecture 10** (Full-line decomposition). *The modulus map  $|\cdot| : X \rightarrow \mathbb{R}_+^*$  and the coordinate  $t = \log |\cdot| : X \rightarrow \mathbb{R}$  induce a decomposition*

$$L^2(X, d^*x) \cong L^2(\mathbb{R}, dt) \widehat{\otimes} L^2(C_K),$$

where  $C_K = \ker(|\cdot|) \subset X$  is the compact fiber (the finite adèlic component at modulus 1), and  $D$  acts as  $-i d/dt \otimes 1$  on this decomposition.

**Theorem 11** (Self-adjointness conditional on decomposition). *If Conjecture 10 holds, then  $D$  is essentially self-adjoint on  $\mathcal{H} = L^2(X, d^*x)$ .*

*Proof.* If  $\mathcal{H} \cong L^2(\mathbb{R}, dt) \otimes L^2(C_K)$  and  $D = D_{\mathbb{R}} \otimes 1$ , then  $\mathcal{K}_{\pm}(\mathcal{H}) = \mathcal{K}_{\pm}(L^2(\mathbb{R})) \otimes L^2(C_K)$ . By Theorem 5,  $\mathcal{K}_{\pm}(L^2(\mathbb{R})) = \{0\}$ . Therefore  $\mathcal{K}_{\pm}(\mathcal{H}) = \{0\}$ , and  $D$  is essentially self-adjoint.  $\square$

**Corollary 12** (RH conditional on decomposition). *If Conjecture 10 holds, and if the zeros of  $\zeta(s)$  appear as  $\rho = \frac{1}{2} + i\lambda$  in the spectrum of  $D$ , then all non-trivial zeros of  $\zeta(s)$  satisfy  $\text{Re}(\rho) = \frac{1}{2}$ .*

## 5 The Gap

*Remark 13* (What remains). The mathematics of Theorem 5 is complete. The gap is Conjecture 10: the decomposition of  $L^2(X, d^*x)$  as a full-line  $L^2$  space.

The difficulty is at two points:

1. **The zero element.**  $X = \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^*$  contains the class of 0. The modulus  $|0| = 0$  sends  $t = \log |0| = -\infty$ . The behavior of functions in  $L^2(X, d^*x)$  near this point must be controlled. This is the archimedean boundary condition at  $-\infty$ .
2. **The Haar measure.** The multiplicative Haar measure  $d^*x$  on  $X$  must decompose as  $dt \times d\mu_{C_K}$  under the product structure  $\mathbb{R} \times C_K$ . The Fourier analysis on  $C_K$  (the compact fiber) must not introduce additional  $L^2$  solutions to the deficiency equations.

Both reduce to archimedean estimates: the behavior of  $L^2(X, d^*x)$  functions at  $t \rightarrow \pm\infty$  in the log-coordinate. This is the Wall. It is archimedean. It has always been archimedean.

## References

- [1] Rincón, D. (2026). Week 09: The Resonance Object. Preprint.
- [2] Rincón, D. (2026). Week 10: Self-Adjointness. Preprint.

- [3] Connes, A. (1999). Trace formula in noncommutative geometry and the zeros of the Riemann zeta function. *Selecta Mathematica* 5(1), 29–106.
- [4] Reed, M. and Simon, B. (1975). *Methods of Modern Mathematical Physics II: Fourier Analysis, Self-Adjointness*. Academic Press.
- [5] Rincón, D. (2026). The Arithmetic Wall. Preprint.